Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/27/2007
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Project Title:	Effect Of Gravity On DNA Transactions In Mammalian Cells		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Waldman, Barbara (University of South Carolina)		
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Genetic recombination plays fundamental roles in many biological processes including regulation of gene expression, DNA repair, disease etiology, and evolution. For over a decade, we have used an experimental system to study intrachromosomal recombination in mammals. We introduce DNA substrates containing two different defective herpes simplex virus thymidine kinase (tk) gene sequences into mouse L cells that are tk-deficient. The pair of defective tk sequences resides as a closely linked direct repeat in the genome. Cells lines are expanded and plated into selective medium in order to recover tk-positive segregants that had undergone a recombination event. From the number of tk-positive segregants that arise, we calculate the rate of recombination in terms of number of events per cell per generation. We also analyze recombination products at the nucleotide level. Recently, we have devised a scheme for studying DNA double-strand break-induced recombination. We now propose to us our model system to study the effects of varying gravitational force on spontaneous and damage-induced recombination in mammals. We will use the Hypergravity Facility for Cell Culture to examine the effects of hypergravity of 4g and we will use a Rotating Cell

Task Description:	Culture System and a clinostat to study the effects of simulated microgravity. The following questions will be addressed: 1.What influence does altered gravitational force have on the rate of intrachromosomal homologous recombination and the relative proportions of reciprocal versus nonreciprocal genetic exchanges in mammalian chromosomes? 2.What influence does altered gravitational force have on the types of mutations that may be induced at DSBs in mammalian chromosomes? 3.What influence does altered gravitational force have on the sensitivity of intrachromosomal recombination in mammalian chromosomes to different degrees and lengths of sequence identity? 4.What influence does altered gravitational force have on the amount of genetic information exchanged during recombination in mammalian chromosomes? We hope to better our understanding of the role of the genome in sensing and responding to gravitational force, to increase our understanding of the possible role of gravity in evolution, and to gain important knowledge relevant to the risks faced by humans in the space environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our project provides a further understanding of the basic processes of DNA recombination and DNA repair and how they may be perturbed. By enhancing our understanding of such DNA transactions, we are gaining valuable information about cellular mechanisms that help to maintain genome integrity which, when compromised, can lead to genetic disease such as cancer.
Task Progress:	We have completed a total of four experiments using the HyFaCC centrifuge at the NASA Ames Research Center (Palo Alto, CA) to study the impact that growth of cells at 4g has on homologous recombination in mammalian chromosomes. Two independently derived mouse cell lines containing an integrated recombination reporter construct were used in these studies. The rate of recombination in all cultures grown at 4g as well as 1g control cultures grown at Ames and 1g control cultures maintained in our laboratory in South Carolina was approximately 1event per cell per million generations. In addition, three experiments were conducted using a 1 foot centrifuge at NASA Ames Research Center to study the effect that cell growth at 10g has on recombination. Again, we observed no significant effect of hypergravity on the rate of intrachromosomal recombination, which was again roughly 1 event per cell per million generations in all cases. A total of 538 recombination events recovered from such experiments (including the control experiments in which cells were maintained at 1g) were analyzed by a combination of Southern blotting, PCR analysis, restriction analysis, and DNA sequencing. From these analyses, we were able to categorize each recombination event as a gene conversion or a crossover. For gene conversions, we could further determine which of the two closely linked interacting marker genes in the reporter construct had undergone conversion. We observed no alteration in the spectrum of the types of recombination events recovered under 1g versus 4g or 10g. We did note a preponderance of gene conversion events involved correction of one marker gene in particular, and we also observed several complex recombination events (involving multiple recombination events) among the large number of samples analyzed. However, there was no apparent effect of altered gravity on the events we recovered and analyzed. In addition to the hypergravity experiments, we also conducted two experiments in our lab in which we examined the effect of simulated microgravity on recombination. To do this, we cultured cells under constant slow rotation around an axis that was perpendicular to Earth's gravity vector. This created a condition of "vector- averaged gravity" in which the orientation of individual cells with respect to Earth's gravity was constantly changing. This resulted, to a first approximation, in a vector sum of essentially zero gravity. Under such growth conditions, we observed no effect on recombination rate relative to cells maintained under control growth under conditions of 1g. Further, analysis of a total of 80 recombination events recovered revealed no alteration in the spectrum of the types of events recovered under 1g versus microgravity. Four trials were also conducted in which the impact of vector-averaged gravity (microgravity) on DNA double-strand break (DSB) repair was assessed. To accomplish this, cells grown under constant slow rotation were briefly removed from the rotation apparatus and an endonuclease was introduced into the cells by electroporation of an expression plasmid in order to induce a defined DSB in the integrated recombination substrate. Cells were returned to the microgravity environment for several days, and then plated into appropriate medium to select for break-repair events arising from recombination or other more error-prone pathways. From these experiments, we observed a statistically significant impact of microgravity on the frequency of DSB break repair events recovered. For control cells grown at 1g, the average frequency of recovery of DSB repair events was 8.71 x 10^-4 events per cell plated into selection . For cells grown under microgravity conditions, the average frequency of recovery of DSB repair events was increased to 1.29 x 10^-3 events per cell plated into selection. We are in the process of analyzing, at the nucleotide level, 144 DSB repair events recovered from cells grown under microgravity and control 1g conditions to learn if the nature of repair is altered under conditions of microgravity . New Technology: NONE A manuscript, "Influence of Alterations in Gravitational Force on Homologous Recombination in Mammalian Chromosomes," is in preparation for submission to submitted to the journal, "Molecular and Cellular Biology."
Bibliography Type:	Description: (Last Updated: 03/20/2017)
Abstracts for Journals and Proceedings	Waldman AS, Wang Y, Waldman BC. "Influence of Altered Gravity on Intrachromosomal Homologous Recombination in Mammalian Cells. " Experimental Biology 2006, San Francisco, CA, April 1-5, 2006. FASEB J. 2006;20: A909. , Apr-2006